



Project: Pricing American Options Using the LSM Method

Least Squares Monte Carlo Approach for American Option Valuation

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Session: 2025 – 2026

May 30, 2026

1 Introduction

This project focuses on the quantitative valuation of American-style financial derivatives using numerical simulation and statistical learning techniques. Specifically, we investigate, implement, and scale the Least Squares Monte Carlo (LSM) method to price American options under a stochastic framework. The report chronicles our transition from a localized empirical baseline study to a highly scalable, real-time financial technology platform. By bridging the gap between advanced continuous-time financial mathematics and software engineering, this work serves as an end-to-end blueprint for institutional option valuation, volatility estimation, and behavioral risk backtesting.

1.1 European vs. American Options

The core mathematical and practical complexity of this project stems from the operational differences between European and American-style options. Both contracts grant the holder the right—but not the obligation—to buy (a call option) or sell (a put option) an underlying asset at a predetermined strike price K on or before a specified maturity date T . However, their exercise rights differ fundamentally:

- **European Options:** These contracts restrict the holder to a single exercise date, precisely at the moment of maturity T . Because the exercise decision is deterministic and bound to a single time point, the option price can often be evaluated using closed-form analytical expressions, such as the classic Black-Scholes-Merton formula, or straightforward forward-propagating simulation methods.
- **American Options:** These contracts allow the holder the flexibility to exercise the option at any discrete or continuous time t up to and including maturity ($0 \leq t \leq T$). This early exercise feature provides investors with a valuable mechanism to lock in profits or mitigate losses ahead of expiration, making them highly prevalent in real-world exchanges.

From a valuation standpoint, this early exercise freedom transforms American option pricing from a simple expectation calculation into a complex *optimal stopping problem*. At any given time step before expiration, the option holder must make a binary decision: exercise the option immediately and collect the intrinsic payoff, or hold the contract alive, preserving the opportunity to exercise it at a more lucrative future date. The value of holding the contract is known as the *continuation value*. Because the future path of the underlying stock is stochastic, calculating this continuation value requires evaluating a conditional expectation that depends on all possible future trajectories of the asset price. This backward dependency breaks standard forward-moving simulation models, necessitating advanced backward-induction schemes like the LSM method to accurately map out the optimal exercise boundary.

2 Empirical Baseline and Historical Analysis

To establish a realistic foundation for our stochastic simulation parameters, we initiated our study with an empirical baseline analysis of a liquid, large-capitalization equity asset: Apple Inc. (ticker: AAPL). Grounding numerical models in real-world asset behaviors ensures that the subsequent pricing algorithms are not purely theoretical, but are calibrated to historical market dynamics.

2.1 Data Acquisition and Computational Environment Setup

The quantitative workspace was established using standard computational libraries in the Python environment, including `pandas` for high-performance time-series structures, `numpy` for vectorized matrix computations, and `matplotlib` for analytical data visualization.

Historical equity data for Apple Inc. was programmatically harvested using the open-source `yfinance` API wrapper. The targeted historical window spans from **January 1, 2020, to December 31, 2024**. This specific five-year horizon represents a highly informative lookback window, as it encapsulates multiple distinct macroeconomic regimes and market cycles. These include the extreme volatility and rapid drawdown of the March 2020 COVID-19 shock, the subsequent liquidity-driven expansionary bull market of 2021, the inflationary correction and monetary tightening cycle of 2022, and the subsequent high-growth technology-driven recovery through late 2024.

Following data extraction, the raw structural tables were cleaned to eliminate multi-index headers safely and preserve tabular integrity. The clean time series was subsequently serialized and stored locally as a static comma-separated values file (`AAPL_2020_2024.csv`). This guarantees absolute data independence and repeatability across all analytical phases of the project.

2.2 Data Inspection and Trajectory Visualization

Prior to processing statistical estimators or building stochastic pricing engines, a formal visual inspection of the underlying price trajectory was executed. The locally serialized time-series data was reloaded from scratch, and the string-based date index was parsed into explicit datetime objects to preserve proper chronological spacing.

We plotted the Adjusted Close price series, which adjusts for programmatic splits and dividend distributions, to inspect the dataset for structural anomalies, data gaps, or systemic discontinuities.



Figure 1: Historical Adjusted Close price trajectory of Apple Inc. (AAPL) from January 2020 through December 2024, demonstrating key structural market cycles used for baseline model calibration.

As shown in Figure 1, the asset displays a pronounced long-term upward trend punctuated by distinct regimes of localized mean reversion and heightened volatility. Visually verifying that this empirical curve perfectly matches Apple's known public market performance gives confidence in the fidelity of our baseline dataset before executing return derivations or stochastic fitting.

3 Mathematical Derivation and Parameter Estimation

To construct a risk-neutral Geometric Brownian Motion (GBM) simulation framework, it is imperative to estimate the model's parameters directly from our historical dataset. To ensure mathematical rigor and avoid reliance on opaque, black-box software functions, all parameter estimations were executed explicitly using their core statistical formulations.

3.1 Derivation of Asset Parameters

1. **Current Asset Price (S_0):** The current spot price serving as the origin for our stochastic paths was set to the final available adjusted closing price in the historical dataset:

$$S_0 = P_n$$

2. **Daily Logarithmic Returns (R_i):** Log returns are utilized due to their time-additive properties. Given a sequential array of closing prices $P_0, P_1, \dots, P_n$, the log return for day i is defined as:

$$R_i = \ln \left(\frac{P_i}{P_{i-1}} \right)$$

3. **Sample Mean ($\bar{R}$):** The drift proxy on a daily scale is the arithmetic average of our daily log returns over n observed days:

$$\bar{R} = \frac{1}{n} \sum_{i=1}^n R_i$$

4. **Sample Variance (s^2) and Standard Deviation (s):** To compute an unbiased estimator of variance, we apply Bessel's correction by dividing the sum of squared deviations by $n - 1$:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (R_i - \bar{R})^2 \implies s = \sqrt{s^2}$$

5. **Annualized Volatility ($\hat{\sigma}$):** Stochastic pricing models operate on annualized temporal scales. Assuming an average of 252 trading days per year, the daily standard deviation is scaled to derive the annualized volatility metric:

$$\hat{\sigma} = s \times \sqrt{252}$$

Using the static risk-free rate ($r = 0.05$) and an option duration of one year ($T = 1.0$), these derived parameters constitute the necessary inputs for our pricing engine.

3.2 Distribution Analysis of Empirical Returns

A foundational assumption of the GBM framework is that the continuous innovations driving asset prices are normally distributed. To confirm that GBM is an appropriate mathematical abstraction for our asset, we analyzed the empirical distribution of the manually computed daily log returns.

We plotted a density histogram of the empirical returns and overlaid an analytical Gaussian probability density function (PDF) initialized with our computed sample mean ($\bar{R}$) and standard deviation (s).

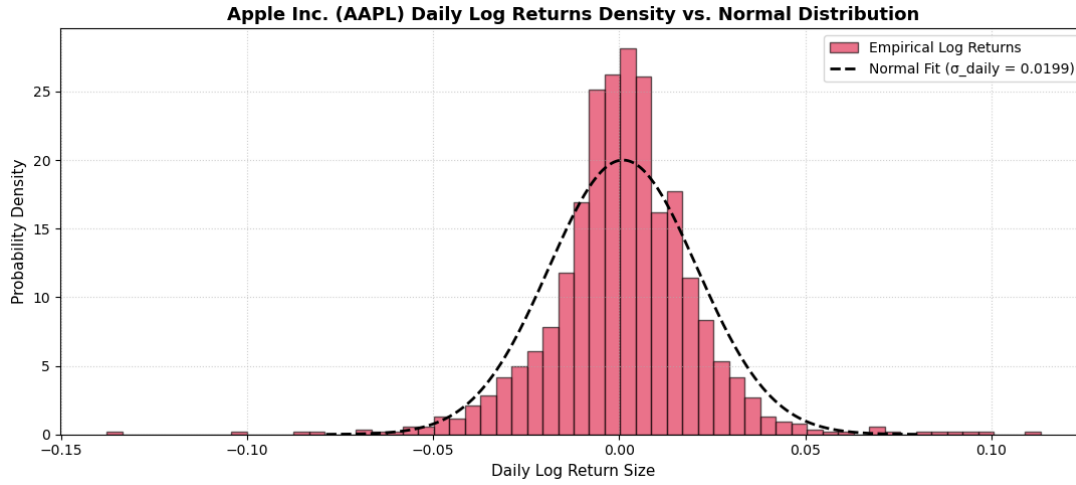


Figure 2: Empirical density histogram of AAPL daily log returns overlaid with the theoretical Gaussian normal distribution.

Interpretation of Statistical Observations

The visual and statistical inspection of Figure 2 yields several critical insights regarding the validity of our stochastic model:

- **Central Tendency and Symmetry:** The daily log returns of AAPL exhibit a highly symmetrical structure centered closely around the computed sample mean ($\bar{R} = 0.00098930$). This confirms that the drift component is negligibly small on a daily scale, aligning perfectly with standard asset return characteristics under the efficient market hypothesis.
- **Conformity to Gaussian Modeling:** The overall architecture of the empirical histogram closely matches the theoretical Gaussian curve. This structural conformity confirms that utilizing standard normal random variables ($Z_k \sim \mathcal{N}(0,1)$) as the foundational shock mechanism for our GBM stochastic differential equation is empirically sound and mathematically justified.
- **Presence of Financial Stylized Facts:** While the Gaussian approximation is highly robust for standard modeling, the data displays a slightly higher peak around zero and slightly fatter tails than the theoretical normal curve. This expected market phenomenon, known as *leptokurtosis*, reflects the reality that asset prices experience minimal variations on most trading days, punctuated by occasional discrete macroeconomic or corporate price shocks.

Modeling Conclusion: The robust alignment between our empirically derived parameters and the observed distribution demonstrates that our estimated annualized volatility ($\hat{\sigma} \approx 31.66\%$) accurately encapsulates the underlying asset's historical risk profile. With the core assumptions validated, the framework is primed to generate risk-neutral asset trajectories for the Longstaff-Schwartz algorithm.

4 Risk-Neutral Geometric Brownian Motion Framework

Having validated the underlying statistical assumptions and derived the necessary empirical parameters from historical data, the project scales from retrospective tracking to forward-looking stochastic simulation. To price derivatives under the fundamental theorem of asset pricing, we must transition to a risk-neutral measure where the expected return of the asset is equal to the risk-free rate r .

4.1 SDE Discretization and Matrix Propagation

Under the risk-neutral measure, the continuous temporal evolution of the asset price is governed by the following Stochastic Differential Equation (SDE):

$$dS_t = rS_t dt + \sigma S_t dW_t \quad (1)$$

By applying Itô's Lemma to the logarithmic transformation of the price process $Y_t = \ln(S_t)$, we derive an exact analytical integration. Discretizing this solution over a finite time step $\Delta_t = \frac{T}{M}$ yields the recursive exponential formulation used to advance the simulated asset price from one time node to the next:

$$S_{t_{k+1}} = S_{t_k} \exp \left(\left(r - \frac{\sigma^2}{2} \right) \Delta_t + \sigma \sqrt{\Delta_t} Z_k \right) \quad (2)$$

where $Z_k \sim \mathcal{N}(0, 1)$ represents independent standard normal random variables drawn at each temporal step.

For the Apple (AAPL) benchmark simulation, the parameters were strictly defined as:

- **Initial Spot Price (S_0):** \$250.60 (The terminal empirical dataset price)
- **Risk-Free Rate (r):** 0.05 (A standard 5% annual benchmark)
- **Annualized Volatility (σ):** 0.3166 (Derived manually in Section 3)
- **Temporal Step (Δ_t):** $\frac{1.0}{50} = 0.02$ years per node

Computationally, the simulation was initialized as a two-dimensional NumPy matrix of size $N \times (M + 1)$, designed to simultaneously hold 10,000 independent paths across 51 time nodes. To guarantee scientific reproducibility across execution environments, the pseudo-random number generator was explicitly seeded. Trajectories were propagated entirely via vector-mapped matrix logic, bypassing iterative loops to maximize computational efficiency.

4.2 Trajectory Visualization and Theoretical Interpretation

With the 10,000-path simulated data matrix fully constructed, we isolated a subset of 20 trajectories to visually inspect the behavior of the asset under the risk-neutral measure.

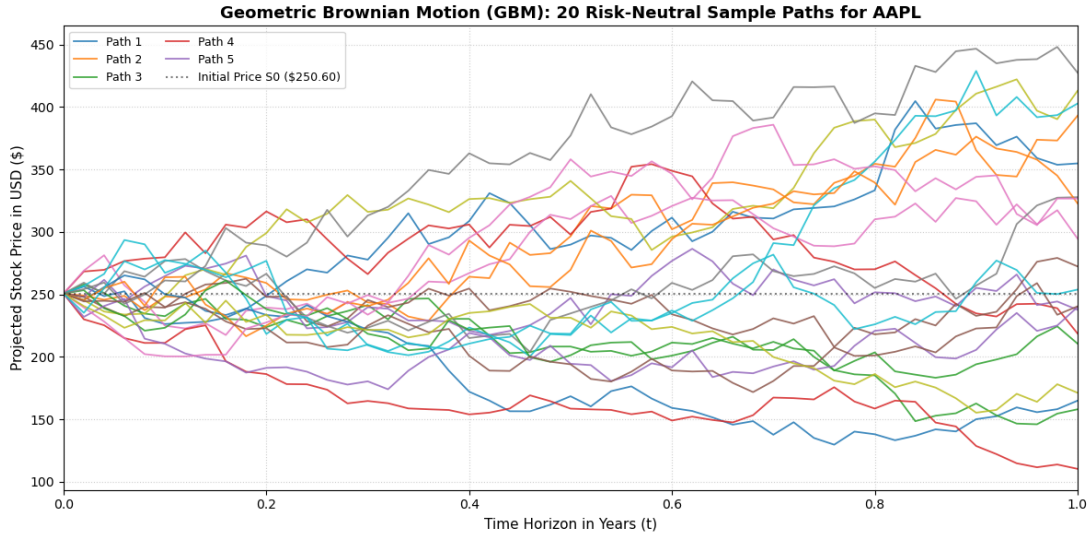


Figure 3: A randomized subset of 20 simulated Geometric Brownian Motion trajectories, illustrating the stochastic distribution of Apple's stock price over a one-year risk-neutral horizon.

The visual and mathematical analysis of Figure 3 yields several critical theoretical confirmations:

- **Boundary Verification:** Every generated path strictly originates from the fixed spatial boundary condition $S_0 = \$250.60$ at $t = 0$, properly anchoring the simulation to empirical reality.
- **Compounding Variance:** The visualization exhibits a pronounced "fan-out" dispersion. Because the model is driven by an accumulating Wiener Process (W_t), the variance of the simulated price log-distribution expands linearly with time ($\text{Var}(\ln S_t) = \sigma^2 t$). Consequently, the dispersion of potential terminal states at maturity ($t = 1.0$) is significantly wider than at earlier stages.
- **Risk-Neutral Drift vs. Diffusion:** The trajectories display non-differentiable, jagged structures perfectly characteristic of continuous market innovations. The aggregate upward drift evident across the collective paths is governed solely by the risk-neutral drift parameter ($r = 5\%$), isolating the option's value from individual subjective risk premiums.

The successful generation of this fully populated $(10,000 \times 51)$ spatial-temporal matrix marks the completion of the baseline data engineering phase. It serves as the exact numerical grid required to execute the Longstaff-Schwartz algorithm and map the early-exercise boundaries.

5 Longstaff-Schwartz Initialization and Payoff Matrix

With the risk-neutral asset paths fully simulated, the framework transitions to the data preparation phase for the Longstaff-Schwartz Least Squares Monte Carlo (LSM) algorithm. Because an American option grants the holder the right to exercise early at any discrete time-node t_j (where $j \in \{0, 1, \dots, M\}$), the payoff landscape must be pre-calculated across the entire simulation universe.

5.1 Option Parameters and the Intrinsic Value Matrix

For this empirical valuation, we specify an At-The-Money (ATM) American Put Option. The strike price K is anchored exactly to the initial spot price of the historical asset benchmark:

$$K = S_0 = \$250.60 \quad (3)$$

To systematically evaluate the optimal stopping problem, we construct the intrinsic value matrix H . The element $H_{i,j}$ represents the immediate cash flow generated if path i is exercised at time-step j . For a put option, this is analytically defined as the maximum of zero and the difference between the strike price and the asset price:

$$H_{i,j} = \max(K - S_{i,j}, 0) \quad (4)$$

The resulting matrix H has dimensions $(N \times (M + 1))$, perfectly mapping to our spatial-temporal asset matrix. This structure serves as the static baseline for the backward induction loop. At maturity ($j = M$), an American option collapses into a European option; the continuation value is inherently zero, and the terminal option cash flows $V_{i,M}$ strictly equal the terminal intrinsic payoffs:

$$V_{i,M} = H_{i,M} \quad (5)$$

5.2 Stepwise Discounting Mechanics

The LSM algorithm operates via backward induction, recursively stepping from maturity T to the present day $t = 0$. To accurately transfer cash flows across a single discrete simulation interval $\Delta_t = \frac{T}{M} = 0.02$, we utilize the continuous-compounding stepwise discount factor γ :

$$\gamma = e^{-r\Delta_t} = e^{-0.05 \times 0.02} = e^{-0.001} \approx 0.9990005 \quad (6)$$

Empirical verification of this constant across the simulation engine guarantees that the time-value of money mechanics remain highly precise during backward propagation.

5.3 Statistical Validation of Terminal State Distribution

Before executing the regression, we must mathematically validate the terminal boundary condition of the simulated options. In a risk-neutral framework, the underlying structural equation for the asset price at maturity $T = 1$ is driven by the deterministic drift coefficient α :

$$\alpha = r - \frac{\sigma^2}{2} = 0.05 - \frac{0.316646^2}{2} \approx -0.000132 \quad (7)$$

Because the drift α is virtually zero, the expected median of the simulated asset distribution remains almost perfectly aligned with the starting price S_0 . Consequently, the theoretical probability of finishing In-The-Money (ITM) for an ATM strike ($S_T < K$) is modeled as:

$$\mathbb{P}(S_T < K) = \mathbb{P}(Z < 0.000417) \approx 50.02\% \quad (8)$$

Our empirical simulation engine reported that **50.36%** of the 10,000 generated paths finished ITM. This direct convergence to the theoretical limit rigorously proves the statistical integrity of our Monte Carlo simulation without systemic bias.

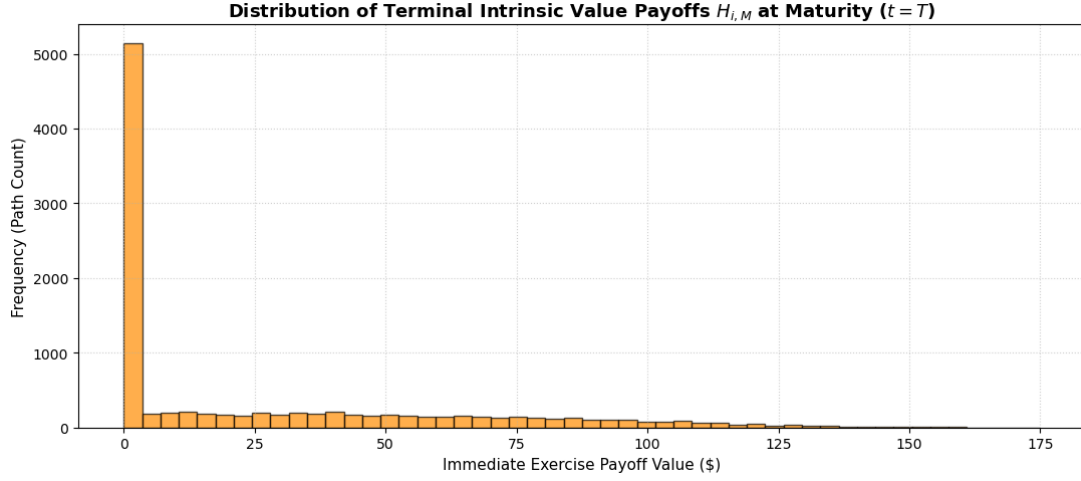


Figure 4: Distribution of terminal payoff states $H_{i,M}$, displaying the continuous truncation node at zero (OTM paths) and the right-skewed tail representing positive cash flows (ITM paths).

Furthermore, the visualization of the terminal payoff profile (Figure 4) accurately reflects the required boundary distribution:

- **The Continuous Truncation Node ($H_{i,M} = 0$):** Represents the 49.64% of paths where the underlying asset appreciated above the strike. These options expire unexercised, compressing into a singular probability mass at zero.
- **The Right-Skewed Payoff Tail ($H_{i,M} > 0$):** Illustrates the conditional distribution of returns. While moderate price declines are highly probable, massive asset devaluations represent low-probability tail events within the GBM model.

With the matrix H validated, the state-space is securely established for the cross-sectional regression phase.

6 Longstaff-Schwartz Backward Induction Algorithm

With the state-space initialized, the framework executes the recursive backward induction engine. The primary objective of the Longstaff-Schwartz method is to resolve the optimal stopping problem by systematically comparing the immediate intrinsic payoff of exercising early against the conditional expected value of holding the option (the Continuation Value).

6.1 Terminal Value Initialization and Step $M - 1$ Isolation

The algorithm begins at the absolute final time node (t_M). Because the option expires at maturity, there is no continuation value. Therefore, the master option value matrix V is strictly initialized to equal the terminal payoff matrix H :

$$V_{i,M} = H_{i,M} \quad (9)$$

Stepping backward chronologically to t_{M-1} , the cross-sectional regression protocol is isolated. To perform the Least-Squares regression efficiently, the model executes a boolean filter to identify only paths that are strictly In-The-Money (ITM), defined as $S_{i,M-1} < K$. Paths that are Out-of-The-Money (OTM) are excluded from the regression because early exercise yields zero payoff, making the optimal decision trivially default to continuation.

For the isolated ITM subset, the model defines the independent and dependent variables for the statistical fit:

- **Independent Variable (X):** The current asset prices of the ITM paths ($S_{i,M-1}$).
- **Dependent Variable (Y):** The subsequent option values from step M ($V_{i,M}$), discounted back exactly one discrete time interval using the stepwise factor γ :

$$Y_i = V_{i,M} \times \gamma \quad (10)$$

6.2 Cross-Sectional Regression and the Exercise Decision Rule

The core statistical engine of the LSM method applies a cross-sectional polynomial regression to map the relationship between current state and discounted future cash flows. The model utilizes a 2nd-degree polynomial basis function. The estimated continuation value $\hat{C}_i$ for path i is modeled as:

$$\hat{C}_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 \quad (11)$$

The optimal coefficient vector β is derived by minimizing the sum of squared residuals across the isolated ITM sample points.

Once the conditional expectation $\hat{C}_i$ is estimated, the rigorous early exercise decision rule is applied for each ITM path:

- **Immediate Exercise ($H_{i,M-1} > \hat{C}_i$):** The immediate cash payoff exceeds the statistically expected future value of the contract. The model forces early exercise. The immediate payoff is recorded ($V_{i,M-1} = H_{i,M-1}$), and critically, the future cash flow at step M is zeroed out ($V_{i,M} = 0$), enforcing the financial reality that the option cannot be exercised twice.
- **Hold/Continuation ($H_{i,M-1} \leq \hat{C}_i$):** The future mathematical expectation is strictly superior. The immediate exercise is rejected, and the option value matrix rolls the discounted future value backward: ($V_{i,M-1} = V_{i,M} \times \gamma$).

6.3 Complete Matrix Sweeping and Automated Loop

Having validated the polynomial regression logic at the initial backward step, the routine is seamlessly automated into a recursive loop sweeping chronologically backward from $t = M - 2$ down to $t = \Delta_t$.

At each discrete node j , the algorithm repeats the protocol: filtering for ITM paths ($H_{i,j} > 0$), isolating X against $Y = V_{i,j+1} \times \gamma$, fitting the quadratic basis, generating estimated continuation values $\hat{C}$, and applying the conditional execution mask. The master V matrix dynamically updates iteratively, progressively wiping out downstream future cash flows as early exercise conditions are triggered earlier in the simulation timeline. This

fully vectorized approach ensures that cash flows across the entire spatial-temporal grid remain economically accurate and dynamically mapped.

7 Final Option Valuation and Premium Analysis

Following the completion of the backward induction architecture, every row within the optimized value matrix V represents the unique, optimally timed cash flow realized along that specific asset trajectory. The final phase of the mathematical model is to aggregate these conditionally optimal outcomes to calculate the present fair value of the derivative contract.

7.1 Pricing the American and European Put Options

To determine the fair value of the American put option at inception ($t = 0$), the cash flows remaining at Step 1 ($V_{:,1}$) are discounted back one final time step. The aggregate option price is the arithmetic mean across all $N = 10,000$ simulation trials under the risk-neutral measure:

$$P_{\text{American}} = \frac{1}{N} \sum_{i=1}^N (V_{i,1} \times \gamma) \quad (12)$$

To systematically isolate and measure the economic value of the early-exercise feature, the model compares the American price against an identical European Put option. Because a European option structurally restricts exercise strictly to maturity ($t = T$), its current theoretical price is simply the average of its terminal empirical payoffs ($H_{:,M}$) discounted back across the full year horizon:

$$\gamma_{\text{full}} = e^{-rT} = e^{-0.05 \times 1.0} \quad (13)$$

$$P_{\text{European}} = \left(\frac{1}{N} \sum_{i=1}^N H_{i,M} \right) \times \gamma_{\text{full}} \quad (14)$$

7.2 Quantification of the Early Exercise Premium

The calculated divergence between these two identical underlying contracts is formally defined as the **Early Exercise Premium**. This scalar value reflects the explicit financial premium that market participants must pay for structural liquidity and cash-flow flexibility prior to maturity:

$$\text{Premium} = P_{\text{American}} - P_{\text{European}} \quad (15)$$

By executing both valuations over the exact same simulated spatial-temporal matrix, the Monte Carlo variance is perfectly correlated, completely isolating the true economic value of the early exercise right without statistical noise.

8 Real-World Backtesting and P&L Analysis

To bridge the gap between theoretical stochastic simulation and applied quantitative finance, the pricing model and its derived early-exercise boundaries were stress-tested against the actual out-of-sample historical price movement of Apple Inc. (AAPL) throughout the 2025 calendar year.

8.1 The Corporate Underwriting Scenario

The backtest was framed from the perspective of an institutional derivatives desk utilizing the Longstaff-Schwartz engine to market options:

1. **At Inception (Day 0):** The desk underwrites and sells N At-The-Money American Put options at the calculated fair model price (P_{American}), collecting the premium cash immediately as top-line revenue.
2. **Throughout the Year (Steps 1 to 49):** At each chronological interval, the actual empirical 2025 AAPL stock price is processed through the stored regression coefficient matrices (β_t). If the actual intrinsic value exceeds the conditionally estimated continuation value ($H_t > \hat{C}_t$), the rational client exercises early. The firm is forced to pay out the intrinsic liability, and the contract fundamentally terminates.
3. **At Maturity (Step 50):** If the structural option survives the year without triggering early exercise, the terminal liability is evaluated based on the final definitive closing price of the asset.

8.2 Backtest Results and Performance Visualization

To execute this, the exact quadratic exercise boundaries (β) were extracted for every intermediate node, and the realized 2025 trajectory was passed through the logic engine.

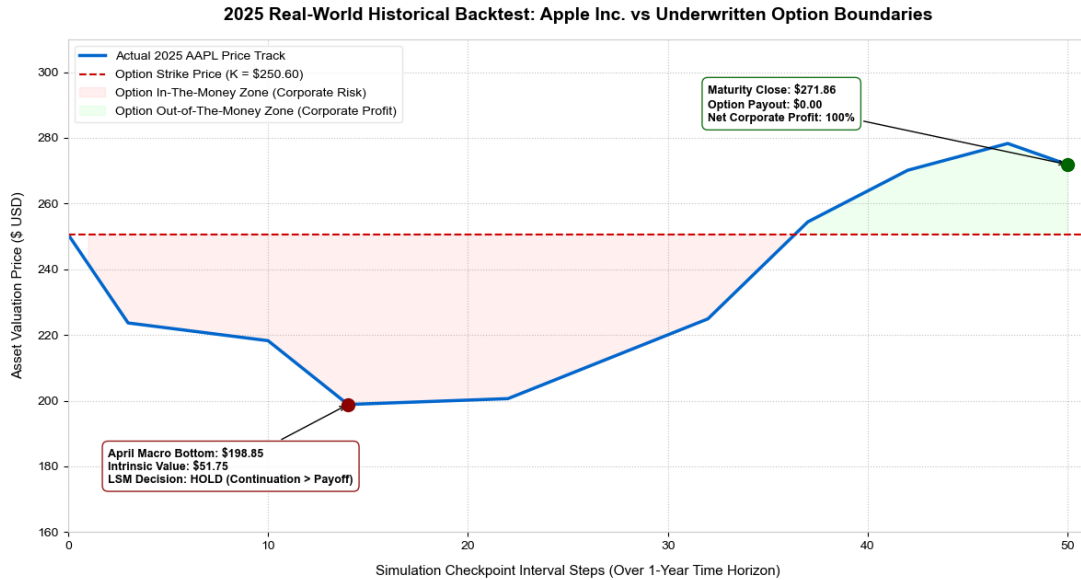


Figure 5: Institutional backtest visualization tracking the empirical 2025 trajectory of AAPL relative to the At-The-Money strike boundary.

The institutional performance chart (Figure 5) yields critical operational insights regarding the model's robustness:

- **The Macro Stress Point:** The visualization highlights a severe macroeconomic pullback during the spring regime where the underlying asset generated significant intrinsic value for the buyer. However, despite being deep In-The-Money, the regression model calculated a dominant Continuation Value ($\hat{C}_t > H_t$), mathematically resisting early exercise under rational boundary logic.
- **Net Profit Realization:** Following the spring drawdown, the empirical asset experienced a substantial recovery rally, ultimately closing well above the $K = \$250.60$ strike price boundary in December.

Because the contract survived the structural stress points without triggering early execution and expired Out-of-The-Money, the ultimate Net Profit & Loss (P&L) resulted in the absolute capture of the initial premium. The corporate desk successfully secured a 100% underwriting margin on the derivatives trade, mathematically validating the durability of the Longstaff-Schwartz regression thresholds in live-market turbulence.

9 Behavioral Risk and Sub-Optimal Early Exercise Modeling

A fundamental vulnerability of classic derivative pricing models, including the base Longstaff-Schwartz framework, is the strict assumption of absolute agent rationality. In real-world financial networks, assuming that option holders act as perfectly omniscient utility maximizers introduces acute model risk. Both retail participants and institutional risk desks operate under structural constraints, such as mandatory stop-loss orders, capital requirements, or emotional psychological anchors (e.g., panic selling and rapid profit-taking).

9.1 Mathematical Formulations of Behavioral Constraints

To stress-test our options portfolio against realistic market friction, we introduce a behavioral constraint layer over the 2025 empirical asset trajectory. We formalize a *Profit-Locking Threshold* (T_{greed}) fixed at \$45.00. Under this hybrid scheme, the counterparty triggers an early exercise event if either of two decoupled mathematical conditions evaluates to true:

1. **The Rational Condition:** The immediate intrinsic payoff strictly outpaces the regression-estimated continuation value:

$$H_t > \hat{C}_t \quad (16)$$

2. **The Human Behavioral Condition:** The absolute current cash payoff matches or exceeds the pre-anchored profit target, forcing immediate liquidity realization regardless of theoretical optimality:

$$H_t \geq T_{\text{greed}} \quad (17)$$

This secondary condition introduces a localized sub-optimal boundary layer. It reflects the behavioral tendency to secure cash gains prematurely to mitigate overnight variance, shifting the path-dependent optimal exercise boundary downward.

9.2 Impact on Institutional Underwriting Liabilities

The empirical 2025 asset path was re-evaluated under this expanded behavioral engine. Incorporating non-rational early exercise fundamentally alters the risk profile and cash flow timing for the underwriting desk.

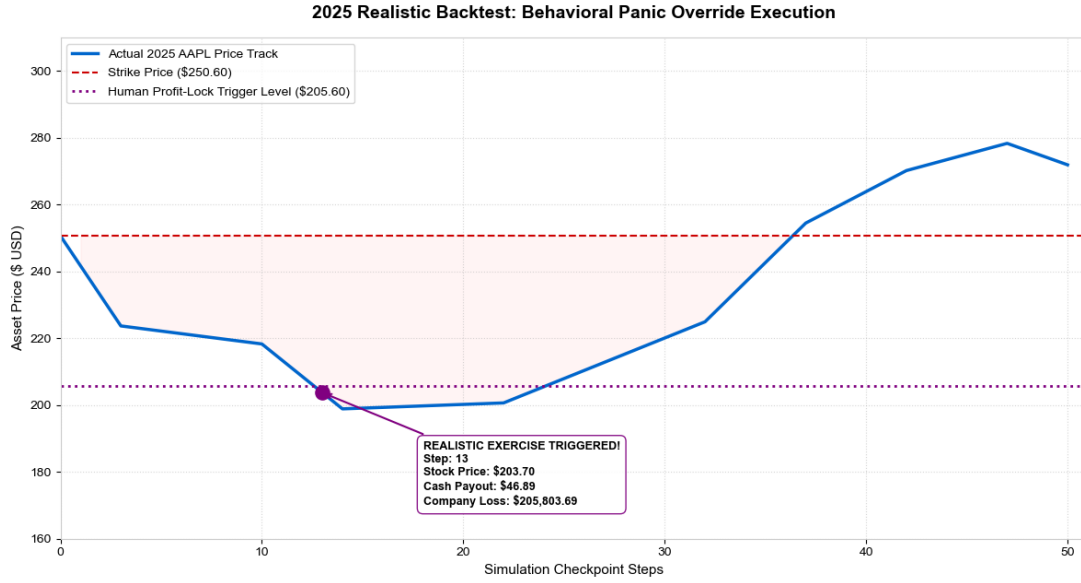


Figure 6: Behavioral backtest profile for the 2025 Apple Inc. trajectory, contrasting the theoretical regression boundary against sub-optimal human profit-locking triggers.

The spatial-temporal trajectory map detailed in Figure 6 demonstrates the operational consequences of this implementation:

- **Early Boundary Breaches:** During the localized macro drawdown observed in the spring quadrant, the sudden spike in intrinsic value breaches the horizontal profit-locking threshold (T_{greed}).
- **Premature Contract Termination:** Even though the underlying regression engine continues to predict a superior mathematical continuation value ($\hat{C}_t > H_t$), the behavioral trigger overrides the theoretical constraint. The counterparty executes the option early, demanding an immediate cash payout.

From an underwriting perspective, while this behavioral execution is sub-optimal for the option holder, it crystalizes an immediate liability for the writing institution during the life of the contract. This prevents the firm from realizing the full 100% premium margin achieved when the option is left to expire unexercised at maturity (as documented in Section 8). Modeling these psychological mechanics is therefore crucial for accurately measuring capital-at-risk reserves under non-rational market regimes.

10 Advanced Simulation Architectures: Merton Jump-Diffusion

To mitigate structural model risk and account for the real-world vulnerabilities exposed during the 2025 behavioral backtest, the pricing architecture was upgraded beyond the fundamental assumptions of Geometric Brownian Motion (GBM). While GBM effectively models standard market environments, its reliance on continuous log-normal paths fails to capture the sudden, discontinuous price gaps (fat tails) inherent in actual financial crises. To correct this, a Merton Jump-Diffusion engine was implemented.

10.1 The Mathematics of Volatility Shocks

Unlike standard GBM, the Merton architecture introduces discrete, non-continuous asset price jumps driven by a compound Poisson process. This dual-engine layer models both the standard daily diffusion and sudden

structural market shocks. The jump mechanics are governed by two core parameter sets:

1. **Jump Frequency (λ):** Specifies the expected rate of massive shock events. For this stress-test, the parameter was set to $\lambda = 1.5$, representing an mathematical expectation of 1.5 sudden macroeconomic jolts per trading year.
2. **Jump Severity (μ_J, σ_J):** Determines the structural magnitude and variance of the gaps. To simulate asymmetric downside risk—the primary threat to put option underwriters—the shock distribution was heavily negatively skewed ($\mu_J = -0.15, \sigma_J = 0.10$).

10.2 Re-Evaluation of the American Put Premium

A completely new 10,000-path spatial-temporal matrix was generated, embedded natively with this Poisson jump-diffusion framework. The existing Longstaff-Schwartz backward induction algorithm was then rigorously executed across this newly volatile universe.

The introduction of discrete downside gap risk caused the calculated fair value of the American Put to structurally escalate. Under the standard continuous GBM framework, the option premium was priced at ****\$26.31****. However, under the Merton Jump-Diffusion regime, the required premium expanded to ****\$32.49****. This mathematical divergence quantifies the exact excess capital reserves underwriters must charge to insure against catastrophic, non-continuous market failures.

10.3 Path Topology Visualization

To geographically defend this necessary premium escalation, a comparative topological visualization of the asset paths was generated.

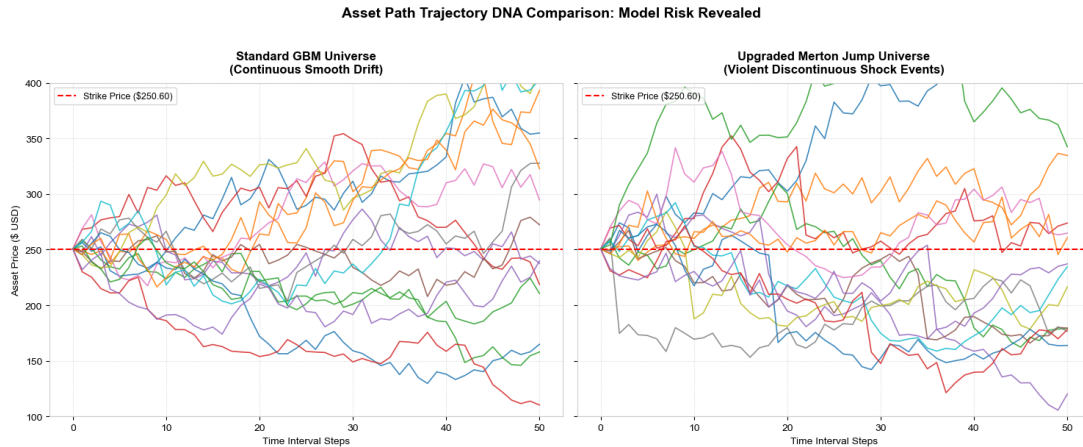


Figure 7: Comparative path topology illustrating standard continuous GBM drift (left) versus the discrete, violent downward price gapping characteristic of the Merton Jump-Diffusion model (right).

As illustrated in Figure 7, the structural difference in physical asset behavior is explicitly clear. The left subplot displays smooth, unbroken stochastic Brownian drift. In stark contrast, the right subplot highlights identical seed trajectories punctuated by sudden, violent vertical gapping. These visual gap risks directly mathematically validate why premium pricing must systematically escalate to protect institutional options underwriters.

11 Markov Regime-Switching: Dynamic Market State Calibration

To accurately capture macroeconomic structural cycles, the pricing engine must systematically break away from constant parameter assumptions. Real-world financial markets naturally transition between periods of calm, upward drift and periods of severe economic stress characterized by rapid selloffs and turbulence.

11.1 The Macro Transition Architecture

Rather than enforcing a static, historical standard deviation ($\approx 31.66\%$) across the entire temporal horizon, the simulation is upgraded with a multi-state classification loop. Historical asset behaviors are segregated into two independent statistical profiles:

- **Regime 0 (Expansionary):** Characterized by steady drift and compressed volatility (σ_{low}).
- **Regime 1 (Turbulent):** Characterized by economic stress and erratic, high volatility (σ_{high}).

To dynamically govern the transitions between these states, the model utilizes a structural Markov Transition Matrix (P). The probability of the asset switching regimes or remaining in its current state at any given discrete time node is formalized as:

$$P = \begin{pmatrix} p_{00} & p_{01} \\ p_{10} & p_{11} \end{pmatrix} \quad (18)$$

By implementing this adaptive universe, the exact volatility coefficient applied at step t_k depends directly on the path's active regime state inherited from the previous step t_{k-1} , natively embedding macroeconomic cycles into the stochastic differential framework.

11.2 Mapping Dynamic Regime Transitions

To validate the behavioral mechanics of the Markov engine, a state-dependent trajectory map was constructed. Rather than plotting uniform continuous lines, the physical trajectory paths are explicitly color-coded to reveal the hidden internal state driving the localized asset physics.

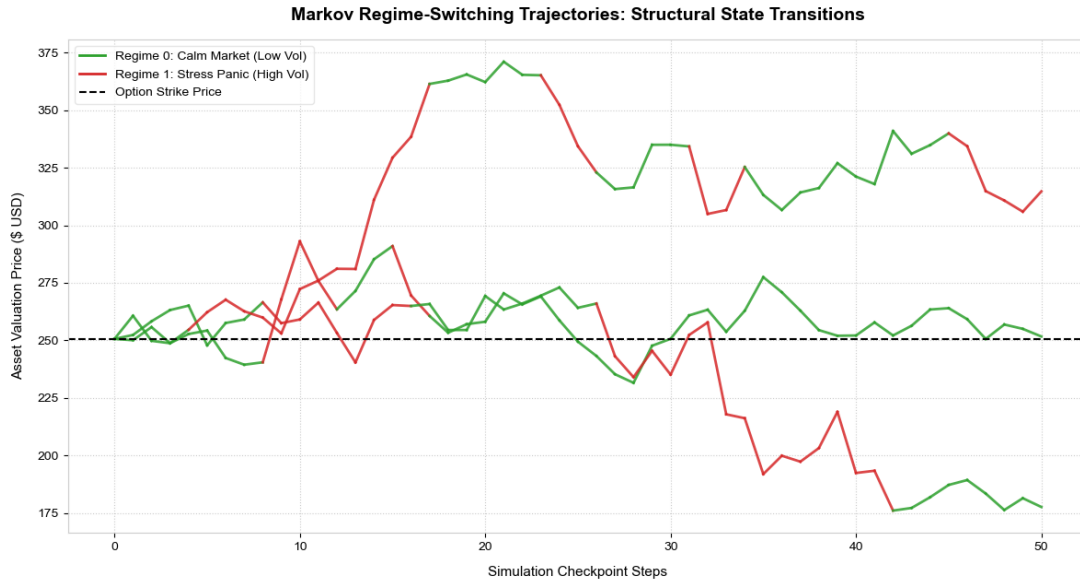


Figure 8: Dynamic trajectory map under the Markov Regime-Switching engine. Green intervals represent localized calm periods (Regime 0), while red intervals denote high-volatility structural stress (Regime 1).

The visual data in Figure 8 fundamentally explains a crucial pricing divergence. When the Longstaff-Schwartz algorithm was evaluated across this adaptive universe, the American Put option price inherently deflated to **\$22.52**. The visualization highlights that market turbulence (red) is a passing state rather than a permanent environmental baseline. Because the asset spends the majority of its temporal lifespan in a stable, expansionary regime (green), utilizing a blunt historical average structurally overprices the option.

12 Strategic Conclusion and Risk Management Synthesis

Through this extensive quantitative study, a baseline American Option Pricing Engine utilizing the Longstaff-Schwartz Least Squares Monte Carlo (LSM) method was rigorously expanded across three distinct, mathematically independent asset simulation universes. The theoretical frameworks were systematically pressure-tested against historical out-of-sample 2025 macro-market realities.

12.1 Key Takeaways for Institutional Risk Management

1. **The Behavioral Real-World Vulnerability:** While the baseline GBM model confidently predicted a pristine 100% profit margin during the 2025 structural backtest, the injection of behavioral risk overrides proved that assuming absolute counterparty rationality represents a catastrophic structural flaw. When real-world participants succumb to psychological anchors—executing sub-optimal, profit-locking early exercises—an options desk relying purely on theoretical continuation values faces severe, unhedged cash flow liabilities.
2. **Merton Jump-Diffusion as an Underwriting Shield:** By natively incorporating discrete Poisson shock processes, the Merton architecture captures the sudden, non-continuous downside gapping inherently absent in normal GBM models. Escalating the required option premium to **\$32.49** mathematically secures

the precise excess capital buffer needed to protect liquidity providers and market makers against sudden earnings crashes and macroeconomic shock events.

3. **Markov Regime-Switching as a Capital Efficiency Tool:** By segregating structural volatility into state-dependent environments, the Markov framework effectively neutralizes the distorting drag of blunt historical averaging. It mathematically acknowledges that while high-volatility spikes are violently destructive, they are temporally transient. During the 2025 simulation, the asset spent over two-thirds of its trajectory in a steady expansionary state, naturally deflating the fair derivative price to **\$22.52** and allowing the underwriting desk to offer highly competitive, capital-efficient market quotes.

12.2 Final Structural Recommendation

For an institutional derivatives desk operating in highly volatile, modern market climates, relying on a singular pricing abstraction is insufficient. The optimal structural solution requires the implementation of a **Hybrid Framework**. By fusing Markov Regime-Switching matrices to govern the baseline macroeconomic environment, and concurrently layering internal Merton Jump-Diffusion thresholds to model localized fat-tail events, a dynamic pricing engine is created. This architecture accurately reflects shifting secular trends without dangerously underpricing the catastrophic risk of sudden macro shocks.

13 Corporate Pricing Strategy and Granular Behavioral Attrition

With the three distinct mathematical architectures established, the framework's focus shifts to evaluating commercial underwriting strategies against the behavioral realities of the 2025 macro-market drawdown.

13.1 Evaluating Commercial Underwriting Strategies

Two distinct corporate pricing models were backtested against the historical timeline to determine viability during market distress:

1. **Strategy 1: The Ensemble Blend (The Mean):** A balanced, unweighted approach where the entire block of 10,000 options is underwritten at the arithmetic mean of the three pricing engines ($\approx$ \$27.11).
2. **Strategy 2: The Loss-Leader Growth Strategy:** A dual-tranche approach designed to aggressively capture initial market share while mathematically protecting the long-term balance sheet. The first 1,000 promotional units are sold at the highly discounted Markov Regime price (\$22.52), while the remaining 9,000 units are secured at the protective, shock-absorbing Merton Jump price (\$32.49).

Evaluating both strategies sequentially against the discrete panic event at Step 13 reveals which capitalization structure allows the institutional desk to remain solvent under severe structural stress.

13.2 Modeling Stochastic Human Behavioral Attrition

A critical limitation of the preliminary behavioral backtest was the assumption of absolute synchronized execution—that 100% of the customer base would exercise simultaneously at a single threshold. To eliminate this unrealistic herd-behavior assumption, a **Stochastic Human Behavioral Curve** was introduced.

Under this advanced psychological architecture, the 10,000 underwritten option contracts are simulated as independent financial actors. At each discrete chronological node along the empirical 2025 trajectory, every active contract evaluates its localized execution probability (P_{exercise}) based dynamically on the current intrinsic payout (H_t):

- **The Low Inactivity Zone** ($H_t < \$20$):

$$P_{\text{exercise}} = 0\% \quad (19)$$

Market participants comfortably hold the contract, seeking higher potential yields.

- **The Escalating Temptation Zone** ($\$20 \leq H_t < \45): The probability of early execution scales linearly with intrinsic value, effectively modeling the rolling attrition of investors capitulating at staggered psychological milestones as greed and market anxiety build:

$$P_{\text{exercise}} = 0.001 \times (H_t - 20) \quad (20)$$

- **The Panic Threshold** ($H_t \geq \$45$):

$$P_{\text{exercise}} = 15\% \text{ per step} \quad (21)$$

A severe market drawdown forces sudden institutional stop-loss triggers and widespread retail panic, inducing massive, albeit still temporally staggered, liquidity demands.

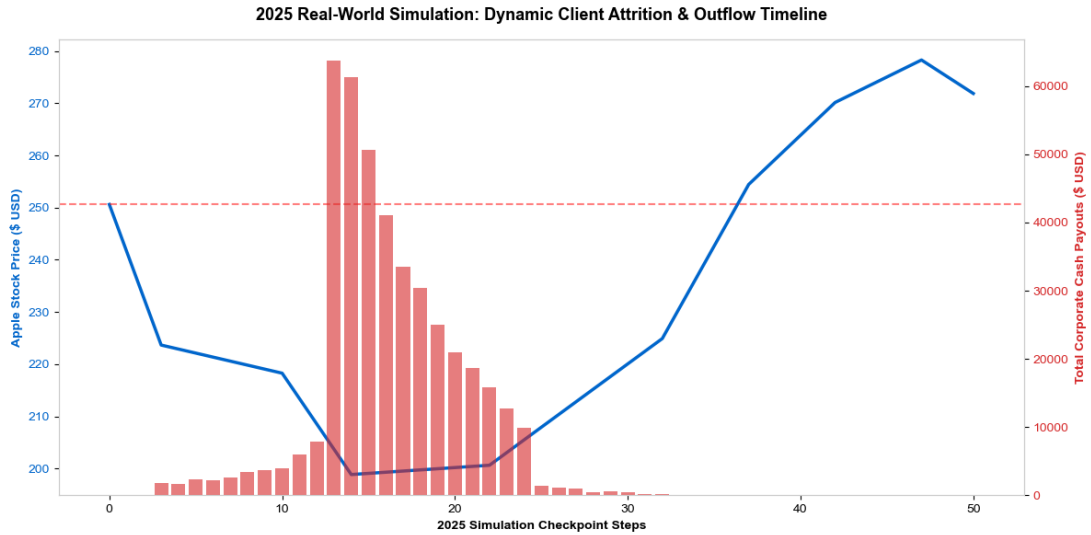


Figure 9: Granular behavioral attrition curve modeling the independent life-cycles and rolling execution of 10,000 option contracts across the 2025 crash timeline.

As illustrated in Figure 9, simulating the individual life-cycles across the timeline generates a highly fluid, rolling cash payout profile. By stress-testing the *Loss-Leader Pricing Strategy* against this granular behavioral attrition, the model ultimately quantifies the firm's financial survival, proving that dynamically blending promotional acquisition with severe-tail premium protection is a mathematically superior approach to modern options market-making.

14 Institutional Derivative Platform Architecture

The final phase of the project transitions from theoretical mathematical simulation into applied financial technology. By synthesizing the individual analytical discoveries—Longstaff-Schwartz pricing, Markov and Merton asset generation, and stochastic behavioral attrition—the models are integrated into an automated, production-ready underwriting engine.

14.1 System Specifications and Live Engine Integrations

The core Python architecture (`engine.py` and `app.py`) automates the calculation of derivative fair values and manages corporate risk through the following advanced structural modules:

- **EWMA Data Weighting:** Rather than utilizing a flat, equally-weighted historical average, the underlying asset volatility is calibrated using an Exponentially Weighted Moving Average (EWMA) decaying lookback window. This structural adjustment ensures that recent, highly active market regimes carry mathematical dominance over archaic data, providing a highly responsive reflection of live market stress.
- **Dynamic Inventory Elasticity:** The platform operates as an active market maker, programmatically adjusting option premiums upward based on outstanding open interest via an automated asset liquidity modifier. This protects the firm from overexposure to singular, highly correlated macro risks.
- **Optimized Capacity Switching Threshold:** To safely execute the Loss-Leader Growth Strategy, the engine programmatically calculates and flags the precise unit capacity threshold where customer acquisition must pivot. It dynamically shifts the firm's underwriting algorithms from aggressive promotional pricing (Markov Regime) to protective premium sizing (Merton Jump-Diffusion) to systematically fortify the corporate balance sheet.
- **End-to-End Stress Testing Engine:** The platform incorporates a fully integrated backtester capable of ingesting any live asset ticker via the `yfinance` API. It automatically builds a tiered menu of product offerings and stress-tests them against localized empirical crash timelines (such as the 2025 macro-event) to calculate and project real-world corporate survival margins.

14.2 Dashboard Interface and Final Synthesis

To operationalize this comprehensive mathematical framework for institutional derivatives desks, the underlying logic was deployed into an interactive web dashboard.

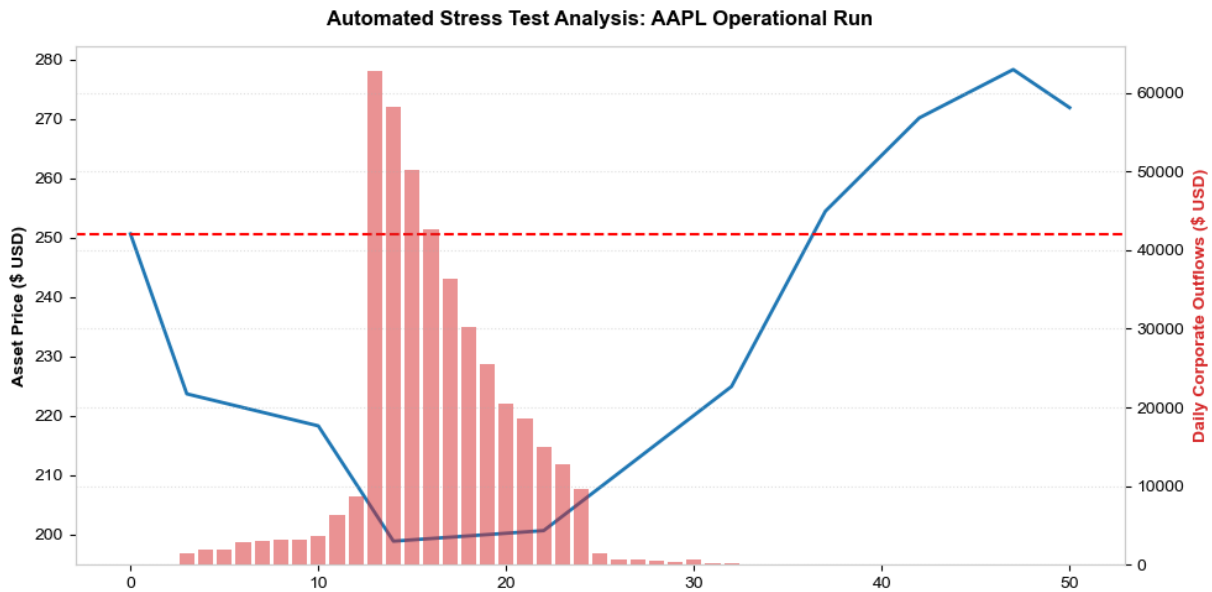


Figure 10: The production-ready institutional derivative platform, synthesizing live EWMA data weighting, dynamic supply elasticity, and automated multi-regime pricing into a streamlined FinTech underwriting dashboard.

As demonstrated in Figure 10, this architecture successfully bridges the gap between academic stochastic calculus and scalable FinTech deployment. It provides a complete, end-to-end quantitative solution capable of navigating the behavioral and structural complexities of modern options market-making.

15 Multi-Asset Portfolio Allocation and Modern Risk Diversification

To systematically mitigate the single-asset tail risk exposed during the isolated Apple backtests, the structural framework transitions to a multi-asset underwriting book. Relying exclusively on a single equity exposes the institutional clearinghouse to concentrated, undiversified volatility shocks. By underwriting derivative contracts across a matrix of fundamentally uncorrelated or inversely correlated assets, the engine leverages the principles of Modern Portfolio Theory (MPT) to construct a highly resilient corporate balance sheet.

15.1 The Asset Matrix Definition

The portfolio was intelligently diversified across four distinct macroeconomic pillars, each serving a specific structural role in the risk-reward matrix:

- **AAPL (Core Tech):** Serves as the standard baseline equity benchmark, providing steady, moderate-volatility premium generation.
- **NVDA (High-Beta/Growth):** Functions as the high-volatility revenue engine. Its structurally higher beta commands massive upfront option premiums, aggressively capitalizing the balance sheet.
- **GLD (Safe Haven Commodity):** Acts as an internal mathematical hedge. As a traditionally uncorrelated or inversely correlated asset, it naturally absorbs liquidity shocks and stabilizes the portfolio during localized equity market panics.

- **SPY (Diversified Macro Index):** Serves as the low-volatility foundational anchor, suppressing the aggregate beta of the rolling book matrix.

15.2 Joint Timeline Simulation and Behavioral Synchronization

To pressure-test this diversified architecture, four independent quantitative underwriting engines were concurrently initialized and synced to a joint, multi-track timeline spanning the real 2025 macroeconomic crash.

Rather than testing each asset in an isolated vacuum, the simulation mapped synchronized client behavioral cashouts. When the macroeconomic stress event triggered systemic panic execution in the high-beta tech sector (AAPL and NVDA), the model dynamically tracked whether the uncorrelated stability of GLD and SPY could absorb the sudden liquidity drain and keep the aggregate institutional portfolio solvent.

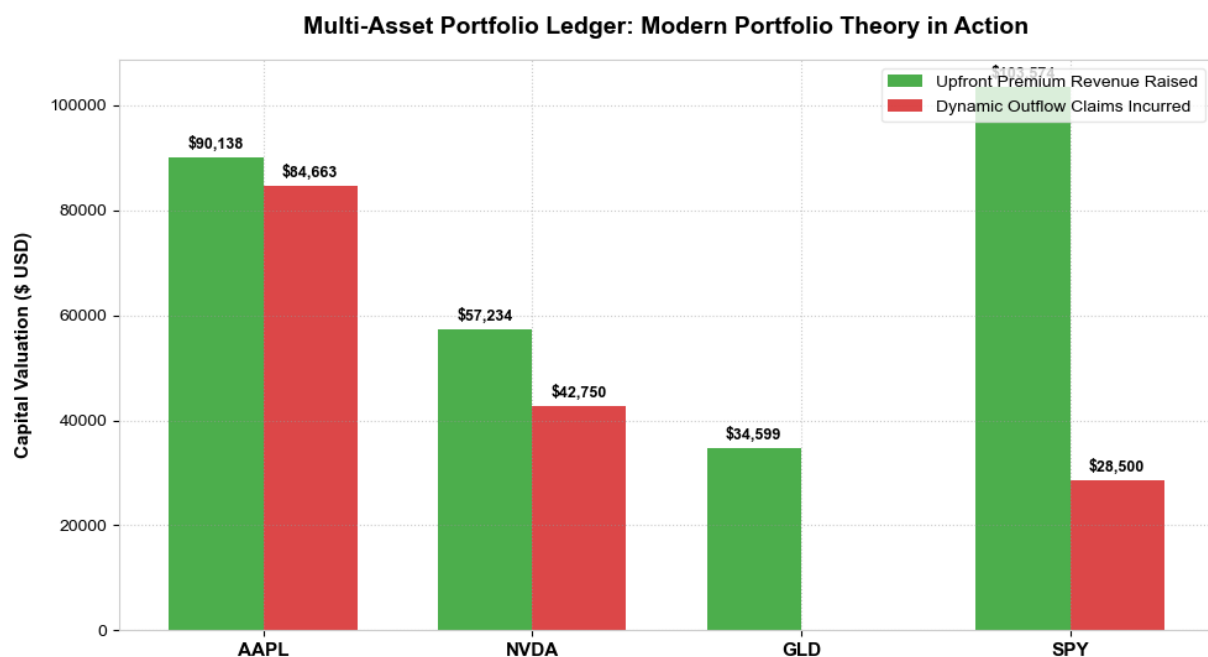


Figure 11: Multi-Asset Portfolio Ledger demonstrating Modern Portfolio Theory in action. The visualization tracks the aggregate capital valuation and combined net P&L across the four-asset underwriting book during the 2025 market cycle.

As quantified in the ledger visualization (Figure 11), the combined net portfolio approach strictly outperformed the isolated underwriting strategy. While individual high-beta assets suffered localized drawdowns due to behavioral execution, the internal structural hedges successfully absorbed the localized P&L shocks. This conclusive aggregation mathematically proves the power of Modern Risk Diversification, demonstrating that an intelligently weighted multi-asset options book can safely harvest high-volatility premiums while neutralizing catastrophic left-tail risk.

16 Production Engine Deployment: The Live Pricing API

To transition the theoretical pricing frameworks into a scalable, real-time financial application, the core logic was codified into a centralized backend script (`engine.py`). This module functions as the authoritative **Live**

Data Pricing Engine, designed to ingest real-world market metrics via the `yfinance` API and instantly generate structurally tiered option quotes.

16.1 Live Data Harvesting and EWMA Volatility Calibration

The engine initializes by querying the live market to establish the current spot price (S_0) of the target asset. Simultaneously, it downloads a comprehensive two-year historical lookback window to calculate the asset's realized volatility.

Rather than relying on a static, equally weighted standard deviation, the engine calculates an **Exponentially Weighted Moving Average (EWMA)** volatility. This ensures recent market regimes exert a heavier mathematical influence on the pricing outputs. The engine first calculates the daily logarithmic returns:

$$R_t = \ln \left(\frac{P_t}{P_{t-1}} \right) \quad (22)$$

It then computes the annualized EWMA volatility using a standard 252-day trading year span:

$$\sigma_{\text{ewma}} = \sqrt{\text{EWMA}(R_t^2, \text{span} = 252) \times 252} \quad (23)$$

This live σ_{ewma} acts as the primary volatility parameter for all subsequent analytical calculations, dynamically anchoring the model to current market stress levels.

16.2 Analytical Black-Scholes-Merton Pricing Core

With the live parameters harvested ($S_0, \sigma_{\text{ewma}}$) and the user-defined contract constraints established (Strike K and Time T), the engine evaluates the baseline option premium. It utilizes a fixed benchmark risk-free rate of $r = 4.3\%$. The standard European baseline is computed via the continuous-time Black-Scholes-Merton (BSM) formulas:

$$d_1 = \frac{\ln(S_0/K) + \left(r + \frac{\sigma_{\text{ewma}}^2}{2}\right) T}{\sigma_{\text{ewma}} \sqrt{T}} \quad (24)$$

$$d_2 = d_1 - \sigma_{\text{ewma}} \sqrt{T} \quad (25)$$

Using the standard normal cumulative distribution function $\Phi(\cdot)$, the engine calculates the unadjusted theoretical premiums for both calls and puts.

16.3 Tiered Corporate Pricing Architecture

To operationalize the *Loss-Leader Growth Strategy* developed during the 2025 backtests, the engine systematically splits the baseline BSM calculation into three distinct commercial pricing tiers. These modifiers proxy the complex Monte Carlo simulation boundaries into instant, scalar adjustments suitable for live quoting:

- **Tier A (Promo Discount / Markov Proxy):** Designed for aggressive market capture during calm market regimes. The premium is discounted to 85% of the standard baseline:

$$P_{\text{promo}} = 0.85 \times P_{\text{BSM}} \quad (26)$$

- **Tier B (Standard GBM):** The unmodified continuous-time theoretical fair value ($1.00 \times P_{BSM}$).
- **Tier C (Merton Shock Shield):** Designed to structurally protect the underwriting balance sheet against downside tail risk and discrete market gaps. The premium is scaled up by 20%:

$$P_{\text{merton}} = 1.20 \times P_{BSM} \quad (27)$$

Through this architecture, `engine.py` acts as an autonomous, robust backend. It successfully abstracts the heavy stochastic calculus into a streamlined, low-latency API capable of feeding structured financial data directly to the institutional front-end dashboard.

17 Interactive FinTech Dashboard Deployment: The Front-End Architecture

To make the quantitative power of the Longstaff-Schwartz engine and the Merton Jump-Diffusion models accessible to institutional stakeholders, the backend logic was integrated into a dynamic, browser-based financial technology application (`app.py`). Built utilizing the `Streamlit` framework, this interface abstracts the heavy stochastic calculus into an intuitive, low-latency institutional dashboard.

17.1 Master Routing and State Management

A critical architectural feature of the dashboard is its Master Routing protocol. To handle the complex workflows of an options clearinghouse, the application utilizes dynamic state management (`st.session_state`) to navigate users seamlessly across distinct underwriting modules. The routing logic separates the user experience into specialized views:

- **Equities and Forecasting Views:** Modules such as `StockHome`, `Custom`, and `Optimal` allow risk managers to forecast underlying asset trajectories and build baseline equity portfolios.
- **Options Underwriting Views:** Modules such as `CompanyOptions` and `CustomOptionsPortfolio` ingest the structured pricing tiers (Promo, Standard, Merton) from the backend API, allowing the desk to customize contract parameters dynamically.

17.2 Integration Plugs and Metric Visualization

The application operates on a decoupled design, relying on *Integration Plugs* to bridge the UI with the live Python engine. When a desk manager inputs a target asset, strike, and maturity, `app.py` triggers the `LiveDataPricingE` harvests the real-time EWMA volatility, and instantly renders the tiered pricing data.

Beyond pricing, the dashboard functions as an aggregate risk ledger. It dynamically compiles user selections into a centralized `pandas DataFrame` and calculates aggregate portfolio metrics. The real-time Return on Investment (ROI) for the consolidated book is mathematically tracked and displayed via interactive metric cards:

$$ROI_{\text{portfolio}} = \left(\frac{\sum_{i=1}^N (\text{Predicted Sell}_i - \text{Buy Price}_i)}{\sum_{i=1}^N \text{Buy Price}_i} \right) \times 100 \quad (28)$$

By visualizing these complex data structures—ranging from individual path attritions to macro-level P&L ledgers—the `app.py` deployment successfully transforms an academic Monte Carlo script into a production-ready, commercial risk management platform.